Method of differences

Basic skills

Q1. $a_n = \frac{1}{n+1} - \frac{1}{n+2}$

a) $a_0 = \frac{1}{1} - \frac{1}{2} = \frac{1}{2}$

b) $a_1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$

c) $a_2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$

d) $a_3 = \frac{1}{4} - \frac{1}{5} = \frac{1}{20}$

e) $a_0 + a_1 = \frac{1}{2} + \frac{1}{6}$ $\leftarrow \left(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} = \frac{2}{3}\right)$

$= \frac{2}{3}$

f) $a_1 + a_2 = \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} = \frac{1}{4}$

g) $a_2 + a_3 = \frac{1}{3} - \frac{1}{4} + \frac{1}{4} - \frac{1}{5} = \frac{2}{15}$

h) $a_0 + a_1 + a_2 = 1 - \cancel{\frac{1}{2}} + \cancel{\frac{1}{2}} - \cancel{\frac{1}{3}} + \cancel{\frac{1}{3}} - \frac{1}{4}$

$= 1 - \frac{1}{4} = \frac{3}{4}$

i) $a_0 + a_1 + a_2 + a_3 = 1 - \dots - \frac{1}{5} = \frac{4}{5}$

j) Predict $1 - \dots$ All cancels out $- 0$

$\to 0$ as $n \to \infty$

$$\sum_{n=0}^{N} a_n = 1 - \frac{1}{N+2} \text{ as } N \to \infty$$

$$\to 1 - 0 = 1$$

So $\displaystyle\sum_{n=0}^{\infty} a_n = 1$

Q2. $f(x) = x^2 - 2x + 3$

$$a_n = f(n+1) - f(n)$$

$$= (n+1)^2 - 2(n+1) + 3 - (n^2 - 2n + 3)$$

$$= n^2 + 2n + 1 - 2n - 2 + 3 - n^2 + 2n - 3$$

$$= 2n - 1$$

Q3. $$\frac{5x+11}{(x+1)(x+3)} = \frac{A}{x+1} + \frac{B}{x+3}$$

$A(x+3) + B(x+1) = 5x + 11$ so

$A + B = 5 \quad 3A + B = 11$

so $B = 5 - A \Rightarrow 2A + 5 = 11 \Rightarrow A = 3$

$\Rightarrow B = 2$ so

$$\frac{5x+11}{(x+1)(x+3)} = \frac{3}{x+1} + \frac{2}{x+3}$$

Q4. $$\frac{3x+5}{(x-3)(2x+1)} = \frac{A}{x-3} + \frac{B}{2x+1}$$

so $A(2x+1) + B(x-3) = 3x+5$

so $2A + B = 3$, $A - 3B = 5$

$A = 5 + 3B \Rightarrow 10 + 7B = 3 \Rightarrow B = -1$

$\Rightarrow A = 2$

so $$\frac{3x+5}{(x-3)(2x+1)} = \frac{2}{x-3} - \frac{1}{2x+1}$$

Competency skills

Q5. $$\sum_{i=0}^{N} \frac{1}{(i+1)(i+2)}$$

a) $$\frac{1}{(i+1)(i+2)} = \frac{A}{i+1} + \frac{B}{i+2}$$

$$A(i+2) + B(i+1) = 1$$

$$A+B=0 \quad 2A+B=1$$

$$B=-A \quad \text{so} \quad A=1, B=-1$$

$$\frac{1}{(i+1)(i+2)} = \frac{1}{i+1} - \frac{1}{i+2}$$

b) $$\sum_{i=0}^{N} \frac{1}{(i+1)(i+2)} = \sum_{i=0}^{N} \left(\frac{1}{n+1} - \frac{1}{n+2}\right)$$

by formula

$$f(0) - f(N+1) = 1 - \frac{1}{N+2}$$

$$= \begin{array}{l} 1 - \frac{1}{2} \\ \frac{1}{2} - \frac{1}{3} \\ \frac{1}{3} - \frac{1}{4} \\ \vdots \\ \frac{1}{N} - \frac{1}{N+1} \\ \frac{1}{N+1} - \frac{1}{N+2} \end{array} \quad \begin{array}{l} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{array}$$

$$= 1 - \frac{1}{N+2}$$

* in practice much better to calculate terms

Q6. $\displaystyle\sum_{i=0}^{n} \frac{2}{(2i+1)(2i+3)}$

Partial fractions

$$\frac{2}{(2i+1)(2i+3)} = \frac{A}{2i+1} + \frac{B}{2i+3}$$

$$A(2i+3) + B(2i+1) = 2$$

$$2A + 2B = 0, \quad 3A + B = 2$$

$$A = -B \Rightarrow -2B = 2 \Rightarrow B = -1, A = 1$$

so $$\frac{2}{(2i+1)(2i+3)} = \frac{1}{2i+1} - \frac{1}{2i+3}$$

so $$\sum_{i=0}^{n} \frac{2}{(2i+1)(2i+3)} = \sum_{i=0}^{n}\left(\frac{1}{2i+1} - \frac{1}{2i+3}\right)$$

$$= \frac{1}{1} - \frac{1}{3}$$
$$+ \frac{1}{3} - \frac{1}{5}$$
$$+ \vdots$$
$$+ \frac{1}{2n-1} - \frac{1}{2n+1}$$
$$+ \frac{1}{2n+1} - \frac{1}{2n+3}$$

0

0

0

0

can setit out like

$1 - \frac{1}{3}$

$\frac{1}{3} - \frac{1}{5}$

$\frac{1}{5} - \dots$

to help

$$= 1 - \frac{1}{2n+3}$$

$$= \frac{2n+3-1}{2n+3}$$

$$= \frac{2(n+1)}{2n+3}$$

Q7. $$\sum_{r=1}^{n}\left(\ln\left(\frac{r+1}{r}\right)\right) = \sum_{r=1}^{n}\left(\ln(r+1) - \ln r\right)$$

law of logs

$$= \ln 2 - \ln 1$$
$$+ \ln 3 - \ln 2$$
$$+$$
$$\vdots$$
$$+ \ln(n) - \ln(n-1)$$
$$+ \ln(n+1) - \ln(n)$$

0

0

0

0

$$= -\ln 1 + \ln(n+1)$$
$$= \ln(n+1)$$

Q8. $\displaystyle\sum_{r=1}^{n} \frac{1}{r(r+1)(r+2)}$

$$\frac{1}{r(r+1)(r+2)} = \frac{A}{r} + \frac{B}{r+1} + \frac{C}{r+2}$$

$$A(r^2+3r+2) + B(r^2+2r) + C(r^2+r) = 1$$

$$A+B+C=0$$
$$3A+2B+C=0$$
$$2A=1 \Rightarrow A=\frac{1}{2} \Rightarrow B+C=-\frac{1}{2}$$
$$2B+C=-\frac{3}{2}$$

$$\Rightarrow B=-1 \Rightarrow C=\frac{1}{2}$$

so $\sum_{r=1}^{n} \frac{1}{r(r+1)(r+2)} = \sum_{r=1}^{n} \left(\frac{\frac{1}{2}}{r} - \frac{1}{r+1} + \frac{\frac{1}{2}}{r+2} \right)$

$$= \sum_{r=1}^{n} \left(\frac{1}{2r} - \frac{1}{r+1} + \frac{1}{2(r+2)} \right)$$

$$= \frac{1}{2} - \frac{1}{2} + \frac{1}{6}$$

$$+ \frac{1}{4} - \frac{1}{3} + \frac{1}{8}$$

$$+ \frac{1}{6} - \frac{1}{4} + \frac{1}{10}$$

$$+ \frac{1}{8} - \frac{1}{5} + \frac{1}{12}$$

don't cancel on row

0

0

$\ddots$

$$\frac{1}{2(n-2)} - \frac{1}{n-1} + \frac{1}{2n}$$

$$+ \frac{1}{2(n-1)} - \frac{1}{n} + \frac{1}{2(n+1)}$$

$$+ \frac{1}{2n} - \frac{1}{n+1} + \frac{1}{2(n+2)}$$

0

0

0

left with

$$\frac{1}{2} - \frac{1}{2} + \frac{1}{4} + \frac{1}{2(n+1)} - \frac{1}{n+1} + \frac{1}{2(n+2)}$$

$$= \frac{1}{4} - \frac{1}{2(n+1)} + \frac{1}{2(n+2)}$$

Advanced skills

Q9.

$$\sum_{i=0}^{N}\left(f(i)-f(i+1)\right)=\sum_{i=0}^{N}f(i)-\sum_{i=0}^{N}f(i+1)$$

index change

$$=\sum_{i=0}^{N}f(i)-\sum_{i=1}^{N+1}f(i)$$

$$=f(0)+\sum_{i=1}^{N}f(i)-\left(\sum_{i=1}^{N}f(i)+f(N+1)\right)$$

$$=f(0)-f(N+1)$$

Q10.

$$\frac{r}{(r+1)(r+2)(r+3)}=\frac{A}{r+1}+\frac{B}{r+2}+\frac{C}{r+3}$$

$$A(r^2+5r+6)+B(r^2+4r+3)+C(r^2+3r+2)=r$$

so $A+B+C=0$ $\quad 6A+3B+2C=0$

$5A+4B+3C=1$

$A = -B - C$

so $-B - 2C = 1$, $-3B - 4C = 0$

$B = -2C - 1 \Rightarrow 6C + 3 - 4C = 0$

$2C + 3 = 0$

$C = -\frac{3}{2}$

$B = 2 \qquad A = -2 + \frac{3}{2} = -\frac{1}{2}$

so $\dfrac{r}{(r+1)(r+2)(r+3)} = \dfrac{-1}{2(r+1)} + \dfrac{2}{r+2} - \dfrac{3}{2(r+3)}$

so $\displaystyle\sum_{r=0}^{N} \left(\frac{-1}{2(r+1)} + \frac{2}{r+2} - \frac{3}{2(r+3)} \right)$

start

$= -\frac{1}{2} + \frac{2}{2} - \frac{3}{6}$

$-\frac{1}{4} + \frac{2}{3} - \frac{3}{8}$

$-\frac{1}{6} + \frac{2}{4} + \dots$

$-\frac{1}{8}$

0 0

end

$$-\frac{1}{2(n-1)} + \frac{2}{n} - \frac{3}{2(n+1)}$$

$$-\frac{2}{2n} + \frac{2}{n+1} - \frac{3}{2(n+2)}$$

$$-\frac{1}{2(n+1)} + \frac{2}{n+2} - \frac{3}{2(n+3)}$$

0 0 0

So $-\frac{1}{2} + \frac{2}{2} - \frac{1}{4} - \frac{3}{2(n+2)} + \frac{2}{n+2} - \frac{3}{2(n+3)}$

$$= \frac{1}{4} - \frac{1}{2(n+2)} - \frac{3}{2(n+3)}$$

Q11. a) $\frac{1}{(r-2)(r+1)} = \frac{A}{r+1} + \frac{B}{r-2}$

$Ar - 2A + Br + B = 1$

$A + B = 0, \quad B - 2A = 1$

so $B = \frac{1}{3}$

$A = -\frac{1}{3}$

so $\sum_{r=5}^{n} \frac{1}{(r-2)(r+1)} = \sum_{r=5}^{n} \left(\frac{1}{3(r-2)} - \frac{1}{3(r+1)} \right)$

$= \frac{1}{9} - \frac{1}{18}$

$+ \frac{1}{12} -$ ← issue here

need to wait till

$(r+1) - (r-2) = 3$ more terms

before cancelling

so NO

b) As mentioned get cancelling further on

going three into future

$\sum_{r=5}^{n} \left(\frac{1}{3(r-2)} - \frac{1}{3(r+1)} \right) = \frac{1}{3} \sum_{r=5}^{n} \frac{1}{r-2} - \frac{1}{3} \sum_{r=5}^{n} \frac{1}{r+1}$

$$= \frac{1}{3}\left(\sum_{r=3}^{n-2} \frac{1}{r} - \sum_{r=6}^{n+1} \frac{1}{r} \right)$$ index change

$$= \frac{1}{3}\left(\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \sum_{r=6}^{n-2} \frac{1}{r} - \left(\sum_{r=6}^{n-2} \frac{1}{r} + \frac{1}{n-1} + \frac{1}{n} + \frac{1}{n+1} \right) \right)$$

$$= \frac{1}{3}\left(\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + 0 - \frac{1}{n-1} - \frac{1}{n} - \frac{1}{n+1} \right)$$

$$= \frac{47}{180} - \frac{1}{3(n-1)} - \frac{1}{3n} - \frac{1}{3(n+1)}$$

Technically this is still method of differences, just non-standard

Q12. a) To find $\sum_{i=0}^{n} \frac{2}{4i^2+16i+15}$

we need to factorise denominator

$$4i^2+16i+15 = (2i+3)(2i+5)$$

Then

$$\frac{2}{(2i+3)(2i+5)} = \frac{A}{2i+3} + \frac{B}{2i+5}$$

$$2Ai + 5A + 2Bi + 3B = 2$$

so $2A + 2B = 0$, $5A + 3B = 2$

$A = -B \Rightarrow -2B = 2 \Rightarrow B = -1$

$A = 1$

$$\sum_{i=0}^{n} \frac{2}{4i^2+16i+15} = \sum_{i=0}^{n} \left(\frac{1}{2i+3} - \frac{1}{2i+5} \right)$$

$$= \frac{1}{3} - \frac{1}{5}$$

$$+ \frac{1}{5} - \frac{1}{7}$$

0 0

$$+ \frac{1}{7} - \dots$$

0 0

$$+ \frac{1}{2n+1} - \frac{1}{2n+3}$$

$$+ \frac{1}{2n+3} - \frac{1}{2n+5}$$

$$= \frac{1}{3} - \frac{1}{2n+5}$$

b) As $n \to \infty$ $\quad \frac{1}{3} - \frac{1}{2n+5} \to \frac{1}{3} - 0 = \frac{1}{3}$

So $\displaystyle\sum_{i=0}^{\infty} \frac{2}{4i^2+16i+15} = \frac{1}{3}$